AI + DS +Healthcare Projects -

Exploring healthcare gaps through a data-driven patient-led lens

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Who am I

My name is Ushasree Jakilinki (https://www.linkedin.com/in/ushasreejakilinki/). I bring a robust foundation of over two decades in software development and technical leadership across the Automotive and Mobile sectors, complemented by recent experience as a Software Program Manager. I hold a Bachelor's degree in Computer Science and Engineering, and I have been accepted into the UofM Master's of Applied Data Science program, starting in August 2025. My long-term aspiration is to pursue a Ph.D. in AI and Healthcare. My dedication to healthcare innovation stems from firsthand encounters with systemic challenges in patient safety, chronic disease diagnosis, and critical care delivery – experiences that have profoundly shaped my belief that AI and data science are uniquely positioned to address these gaps. Beyond healthcare, my passions include education (ignited by witnessing its transformative power firsthand) and leveraging my project management experience to drive complex data science initiatives. My core technical background also includes continued interest in the Automotive sector.

From PCOS/GD to T2D : Can Data Help Change the Equation? Or From PCOS/GD to T2D – Predicting Progression, Changing Outcomes

CAN WE IDENTIFY BIOMARKER TRAJECTORIES & LIFESTYLE FACTORS TO PREDICT AND PREVENT T2D ONSET IN PCOS PATIENTS? PROJECT -1

The Problem & Project Vision

Problem

PCOS is a multifactorial condition impacting endocrine, metabolic, cardiovascular, and emotional health.

Many patients experience delayed diagnosis, misdiagnosis, and lack of targeted intervention.

Common manifestations like acne, weight gain, hair loss, and hirsutism significantly impact a woman's confidence and quality of life.

PCOS has a genetic disposition, but lifestyle changes and management are crucial.

Vision

This project aims to explore the longitudinal progression from PCOS to T2D using a data-driven, patient-centered approach.

By leveraging de-identified clinical data, we seek to:
Identify early warning signs and predictive patterns.

Understand the impact of comorbidities (e.g., thyroid, mental health, GI, metabolic).

Analyze lifestyle, medication, and demographic influences on progression.

Integrate and weigh these factors to inform the patient journey.

Why

Why This Analysis Matters: A Personal and 360° View

- I live with Polycystic Ovary Syndrome (PCOS) and have a strong family history of Type 2 Diabetes (T2D).
- My personal health journey has included years of misdiagnoses, frustration, and confusion — but also breakthrough moments through 360° solutions that combined medical care with lifestyle changes. The profound impact on my self-esteem from symptoms like acne, weight gain, hair loss, and hirsutism was a significant personal challenge, often leading to substantial financial costs for management.
- Through this experience, I've discovered the power of comprehensive, data-driven, 360° solutions.
- My story reflects that real, positive health outcomes are possible for millions of underrepresented women with the right data, holistic care, and long-term strategies.

My 360 approach

- 💊 **Medical Support:** I take Metformin to manage insulin resistance.
- 🥗 **Nutrition:** Switched to whole grains, anti-inflammatory foods, and reduced refined sugar.
- ⏱️ **Intermittent Fasting:** Improved energy, reduced cravings, and better glucose control.
- 🧘 **Movement:** Walking, yoga, Zumba and Spinning to improve metabolic and mental health.
- 🧠 **Mental Wellness:** Therapy, mindfulness, and stress management to support hormone balance.
- 📊 **Tracking:** Food reactions, mood, sleep, and symptom patterns over time.
- I have consulted with multiple PCPs, endocrinologists, infertility specialists , gynecologists, Sleep studies and therapists; tried alternative paths like Ayurveda, acupuncture , nutrition and massage; and explored numerous books and videos on the topic.
- My story reflects those of millions of underrepresented women — real outcomes are possible with the right data, care, and long-term strategies.

A pcos patient needs the...

Specialist	Primary Focus
Primary Care Provider (PCP)	Overall coordination, general health monitoring
Endocrinologist	Hormonal balance, metabolic and insulin resistance issues
Gynecologist/Reproductive Endocrinologist	Menstrual and fertility concerns, reproductive management
Maternal-Fetal Medicine (MFM)/High-Risk OB-GYN	High-risk pregnancy care, monitoring for GD and complications
Dermatologist	Skin/hair symptoms: acne, hirsutism, alopecia
Mental Health Provider (e.g., therapist, psychiatrist)	Emotional, psychological support
Integrative/Functional Medicine Doctor	Holistic and complementary care
Registered Dietitian/Nutritionist (including pregnancy expertise)	Dietary and lifestyle coaching, GD management
Diabetes Educator	GD support: glucose monitoring, dietary guidance

Why

PCOS & Type 2 Diabetes: Prevalence & Risk

- **Polycystic Ovary Syndrome (PCOS):**
  - Affects **6-13%** of reproductive-aged women globally.
  - Up to **70%** of affected women remain undiagnosed.
- **High Risk for Type 2 Diabetes (T2D):**
  - Women with PCOS have a **3-4 times higher risk** of developing T2D.
  - Prevalence of T2D in women with PCOS is **1.5% to 12.4%** (often 7.5-10%).
  - **Key Statistic:** Approximately **50%** of women with PCOS may develop T2D by age 40.
- **Individual Healthcare Costs (Per Patient):**
  - **Lifestyle Interventions:** ~$500-$1,000 annually.
  - **Pharmacological Treatments:** ~$1,000-$2,500 annually.
  - **Multidisciplinary Care:** ~$1,200-$2,000 annually.

To summarize I want to answer
Can we identify specific biomarker trajectories that predict T2D onset in PCOS patients X years in advance? or What lifestyle interventions show the strongest statistical association with delaying or preventing T2D in PCOS patients?
Can we identify early signals in PCOS and GDM that tell us who is likely to develop T2D — and when?

Why

PCOS & T2D: Economic Burden & Data Currency

- **Significant Overall Economic Impact (U.S.):**
  - Total annual cost of PCOS-related comorbidities: **Over $15 Billion** (as of 2021 USD).
  - T2D accounts for **19% to 37.5%** of the total economic burden of PCOS.
  - Estimated annual excess cost for T2D attributable to PCOS: **~$1.5 Billion** (2020/2021 USD).
- **For T2D patients (overall, not just PCOS-related):**
  - Average annual medical expenditures are **~$19,736**, with ~$12,022 directly attributable to diabetes (2022 USD).
  - This is 2.6 times higher than for those without diabetes.
- **Other Major Cost Drivers (U.S. Annual Estimates):**
  - Mental Health Disorders: Over **$4.2 Billion** (2021 USD)
  - Menstrual Dysfunction: ~$2.4 Billion (2021 USD equivalent)
  - Infertility Care: ~$0.95 Billion (2021 USD equivalent)
- **Data Currency:** Information primarily drawn from research and reviews published between 2021 and 2025. Cost figures are in 2020, 2021, or 2022 US Dollars, as specified.

Current solutions

- https://diabetes.org/diabetes-risk-test - this predicts risk based on a question are but its static and does not even ask about PCOS or insulin resistance and such.
- We need a system that takes input from user with a lot of data points, history of pcos patients, etc etc. In short use crowd source

What –Requested data -TBD

1. Demographics & Medical History:

- Age at diagnosis (PCOS, T2D, Gestational Diabetes) — *Integer / Date*
- Ethnicity / Race — *Categorical (e.g., coded values)*
- Family history (Diabetes, Heart Disease) — *Boolean / Categorical*
- Physical sighs (Physical signs (e.g., Acne, Hirsutism, Hair loss, Skin tags, Acanthosis Nigricans, Oily skin) — Boolean)
- *Food intolerances : Dairy, Gluten or Soy or something else*
- Medical issues (e.g., Thyroid, High Cholesterol, GI issues, Sleep issues, Irregular periods, Hypertension, Gestational diabetes, Infertility diagnosis) — Boolean / Categorical

2. Clinical Signs & Symptoms:

- BMI, weight history — *Float / Numeric*
- Body measurements (e.g., waist circumference) — *Float / Numeric*
- Carb cravings — *Categorical / Boolean*
- Menstrual irregularities — *Categorical / Boolean*
- Anxiety and depression — *Categorical / Boolean*
- Sleep disturbances — *Categorical / Boolean*
- *Cortisol –Anti inflamotary ALE*

What –Requested data TBD

3. Labs & Diagnostics

- Blood glucose (fasting, HbA1c) — *Float / Numeric*
- Lipid panel (HDL, LDL, triglycerides) — *Float / Numeric*
- Insulin levels — *Float / Numeric*
- Hormonal markers (LH, FSH, AMH, Testosterone) — *Float / Numeric*
- Thyroid panel (TSH, T3, T4) — *Float / Numeric*
- Iron studies (ferritin, iron, TIBC) — *Float / Numeric*

4.Medications & Allergies

- Prescribed medications — *Text / Categorical*
- Over-the-counter medications — *Text / Categorical*
- Allergy history — *Categorical / Boolean*

5.Lifestyle Factors

- Exercise habits (frequency, type: yoga, pilates, cardio) — *Categorical / Text*
- Nutrition (if available) — *Text / Categorical*
- Smoking status — *Boolean / Categorical*
- Alcohol use — *Boolean / Categorical*
- Sleep quality — *Categorical / Boolean*
- Stress management practices — *Text / Categorical*

Anticipated Challenges & Mitigation Strategies

1. Data Availability & Access:

Obtaining a sufficiently large, comprehensive, and longitudinal de-identified dataset.

2. Data Quality & Fragmentation:

Inconsistent data collection, missing values, varying coding standards, and data silos.

3. Regulatory & Ethical Compliance (HIPAA):

Strict privacy regulations requiring rigorous ethical review and data use agreements.

4. Long-Term Tracking & Causality:

Establishing clear causal links between interventions/trajectories and T2D onset over many years.

1. Mitigation for Data Availability:

Partner with health systems/research institutions. Leverage existing de-identified cohorts. Pilot with accessible datasets.

2. Mitigation for Data Quality:

Employ robust cleaning, imputation, and feature engineering. Prioritize standardized data sources. Acknowledge data limitations.

3. Mitigation for Compliance:

Focus on de-identified/anonymized data.Collaborate with legal, compliance, and IRB teams.Adhere to highest security/privacy standards.

4. Mitigation for Tracking & Causality:

Identify strong statistical associations. Utilize time-series analysis. Clearly define model scope/limitations.

Support or help -TBD

What do I need for this project ?	From whom
Endocrinologist knowledge	University of Chicago –PCOS center Dr David A Ehrmann https://profiles.uchicago.edu/profiles/display/38978 UofM/MEND Endo department
Nutrition	Michigan genomics institute – https://aidhi.umich.edu/news/mgi-rfp?rq=whole-genome
PCOS Datasets	Arika will check MEND for pcos and diabetes EOD wednes
Cardiologist –Heart risk	Sharktank for AI DHI –for clinician in Feb 2026
Sleep medicine –How many pcos patients have sleep issues	11am on Monday 27th meeting with Arika
PCOS or Diabetes research –MEND –	IH PI institute for healthcare policy and, Health services research
Depression and anxiety	

Papers

1. https://pmc.ncbi.nlm.nih.gov/articles/PMC10637759/
2. https://www.cedars-sinai.org/blog/pcos-diabetes-heart-disease-stroke.html
3. https://www.webmd.com/diabetes/pcos-and-diabetes
4. https://www.endocrine.org/news-and-advocacy/news-room/2017/researchers-reveal-link-between-pcos-type-2-diabetes
5. https://www.hopkinsmedicine.org/health/wellness-and-prevention/heart-health
6. https://diabetesjournals.org/diabetes/article/70/2/627/39489/Polycystic-Ovary-Syndrome-and-Risk-of-Type-2
7. https://pmc.ncbi.nlm.nih.gov/articles/PMC2795626/#:~:text=All%20the%20manifestations%20of%20the,restitution%20of%20normal%20insulin%20sensitivity.
8. https://www.yalemedicine.org/conditions/polycystic-ovary-syndrome
9. https://womenshealth.labcorp.com/pcos-more-than-just-reproductive-issue
10. https://www.aafp.org/pubs/afp/afp-community-blog/entry/new-diagnostic-option-for-polycystic-ovarian-syndrome.html

Papers -TBD

Topic	paper	When published
Sleep	1. https://pmc.ncbi.nlm.nih.gov/articles/PMC6724486/ 2. https://pmc.ncbi.nlm.nih.gov/articles/PMC5799701/ 3. https://www.askpcos.org/articles/pcos-and-sleep/	2019 Sep 3 2018 Feb August 15th, 2023